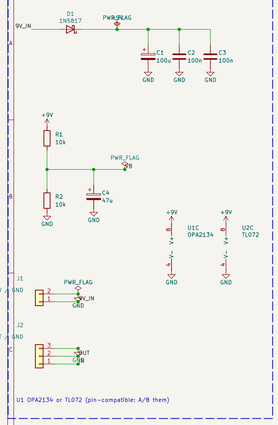
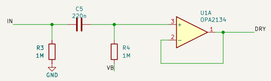
POWER + BIAS + JACKS

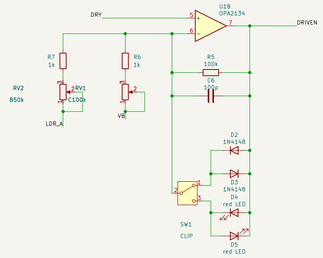


U1 OPA2134 or TL072 (pin-compatible: A/B them)

INPUT BUFFER (U1A): clean signal for the drive stage and the detector



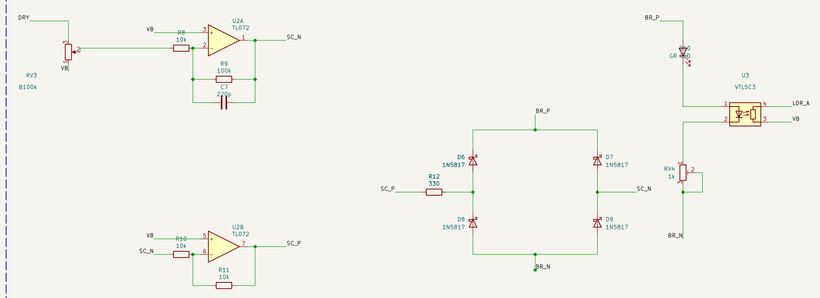
DRIVE STAGE (U1B): gain = 1 + R5 / ((R6 + DRIVE) || (R7 + DEPTH + LDR)); diodes cap the output



DRIVE sets the resting gain (x2 to x101). Picking harder lights the LDR, which pulls the ground leg down through DEPTH: more gain, more clipping. The diodes hold the output at about +-0.6 V (Si) / +-1.8 V (LED), so U1B never clips.

LDR_B goes to VB. Off-board: the pots and switch; everything else fits a breadboard.

DETECTOR (U2): fed from the CLEAN signal through SENSE; full-wave bridge into the vactrol LED



SENSE: how hard you must pick to light the LDR. U2A gain 10. U2B unity inverter: bridge sees 0–20x DRY.

LDR_A goes to the DEPTH leg of the drive stage. LDR_B (pin 3) to VB.

OUTPUT: TONE (hi-cut) + VOLUME

Hard picking clips, soft picking and decays clean up
Optical drive whose gain follows the clean pick envelope (Flatline / Punch Line style)

Super-Freq

Sheet: /
File: CleanUpDrive.kicad_sch

Title: Clean-Up Drive (breadboard)

Size: A1	Date:	**Rev: 0.1**
KiCad E.D.A. 10.0.4		Id: 1/1